

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 3: Comparative Analysis**

**COURSE NAME: COMPUTER VISION**

**COURSE CODE: ITA0510**

**Slot A**

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**COURSE OUTCOME COVERED**

**CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)**

**Assessment Tool Weightage: 10%**

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**QUESTIONS:5**

**Total Marks 50**

**Duration :60 Minutes Annexure A**

**Comparative Analysis**

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**Code of Conduct:**

***I M.THANUJA(192425356) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.***

**Signature of the student: M.THANUJA**

**Rubrics for Evaluation**

| Criteria          | Weightage (%) | Level 4 (Exemplary)                                                           | Level 3 (Proficient)                            | Level 2 (Developing)                       | Level 1 (Beginning)               | Marks(100) |
|-------------------|---------------|-------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------|-----------------------------------|------------|
| Scope of Items    | 20%           | Selects highly relevant items with clear scope and justification              | Selects appropriate items with minor gaps       | Selection is limited or partially relevant | Items are unclear or not relevant |            |
| Comparison Depth  | 25%           | Provides detailed and comprehensive comparison across multiple aspects        | Provides adequate comparison with some depth    | Limited comparison with few aspects        | Very minimal or no comparison     |            |
| Use of Evidence   | 20%           | Supports comparison with accurate data, examples, or references               | Uses some supporting evidence with minor gaps   | Limited or weak supporting evidence        | No supporting evidence provided   |            |
| Critical Thinking | 20%           | Demonstrates strong critical thinking with clear interpretation and reasoning | Shows reasonable interpretation with minor gaps | Basic interpretation; lacks depth          | No meaningful interpretation      |            |

|                         |     |                                                              |                                   |                                       |                                          |  |
|-------------------------|-----|--------------------------------------------------------------|-----------------------------------|---------------------------------------|------------------------------------------|--|
| Clarity of Presentation | 15% | Well-organized, clear, and logically structured presentation | Generally clear with minor issues | Somewhat unclear or poorly structured | Disorganized and difficult to understand |  |
|-------------------------|-----|--------------------------------------------------------------|-----------------------------------|---------------------------------------|------------------------------------------|--|

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

### 1. Detection Performance Comparison

Two object detection models produce the following results:

**Model    True Positives   False Positives   False Negatives**

**A (Viola-Jones)   80      20      30**

**B (YOLO)      90      15      20**

**(a) Compute precision and recall for both models.**

**(b) Compare their performance and state which model is better for object detection.**

**ANS. Given Data**

**Two object detection models are evaluated:**

| Model                 | True Positives<br>(TP) | False Positives<br>(FP) | False Negatives<br>(FN) |
|-----------------------|------------------------|-------------------------|-------------------------|
| Model A – Viola-Jones | 80                     | 20                      | 30                      |
| Model B – YOLO        | 90                     | 15                      | 20                      |

**(a) Compute Precision and Recall for Both Models**

Step 1: Write the formulas

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \times 100$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \times 100$$

Where:

- TP (True Positive): Correctly detected objects
- FP (False Positive): Incorrect detections
- FN (False Negative): Actual objects that were missed

**Step 2: Calculate Precision of Viola-Jones**

For Viola-Jones:

$$TP = 80$$

$$FP = 20$$

$$\text{Precision} = 80 / (80 + 20) \times 100$$

$$= 80 / 100 \times 100$$

$$= 80\%$$

Therefore:

$$\text{Precision of Viola-Jones} = 80\%$$

### **Step 3: Calculate Recall of Viola-Jones**

For Viola-Jones:

$$TP = 80$$

$$FN = 30$$

$$\text{Recall} = 80 / (80 + 30) \times 100$$

$$= 80 / 110 \times 100$$

$$= 72.73\%$$

Therefore:

$$\text{Recall of Viola-Jones} = 72.73\%$$

### **Step 4: Calculate Precision of YOLO**

For YOLO:

$$TP = 90$$

$$FP = 15$$

$$\text{Precision} = 90 / (90 + 15) \times 100$$

$$= 90 / 105 \times 100$$

$$= 85.71\%$$

Therefore:

$$\text{Precision of YOLO} = 85.71\%$$

### **Step 5: Calculate Recall of YOLO**

For YOLO:

TP = 90

FN = 20

Recall =  $90 / (90 + 20) \times 100$

=  $90 / 110 \times 100$

= 81.82%

Therefore:

Recall of YOLO = 81.82%

### Step 6: Compare the Results

| Performance Measure | Viola-Jones | YOLO   |
|---------------------|-------------|--------|
| Precision           | 80%         | 85.71% |
| Recall              | 72.73%      | 81.82% |

### Step 7: Evaluation

YOLO has a higher precision of 85.71% compared with 80% for Viola-Jones. This means YOLO produces fewer false detections.

YOLO also has a higher recall of 81.82% compared with 72.73% for Viola-Jones. This means YOLO detects more of the actual objects in the image.

The improvement in precision is:

$85.71 - 80 = 5.71$  percentage points

The improvement in recall is:

$81.82 - 72.73 = 9.09$  percentage points

YOLO also has fewer false positives (15 vs. 20) and fewer false negatives (20 vs. 30).

### Final Conclusion

YOLO is the better object detection model. It achieves both higher precision (85.71%) and higher recall (81.82%) than Viola-Jones (80% precision and 72.73% recall). Therefore, based on the calculated performance measures, YOLO provides more accurate and reliable object detection for the given data.

## 2. Motion Estimation Error Analysis

**Two motion estimation techniques produce displacement errors (in pixels):**

**Frame Method A Method B**

**1      2      3**

**2      3      5**

**3      2      4**

**(a) Compute average error for both methods.**

**(b) Compare and evaluate which method is more accurate.**

ANS. Given Data

Two motion estimation techniques produce the following displacement errors:

| Frame | Method A | Method B |
|-------|----------|----------|
| 1     | 2 pixels | 3 pixels |
| 2     | 3 pixels | 5 pixels |
| 3     | 2 pixels | 4 pixels |

**(a) Compute Average Error for Both Methods**

Step 1: Write the formula

Average Error = Total Error / Number of Frames

A lower average error indicates better motion estimation accuracy.

**Step 2: Calculate Average Error for Method A**

Method A errors are:

2, 3, and 2 pixels.

First, add the errors:

$2 + 3 + 2 = 7$  pixels

Number of frames = 3

Average Error =  $7 / 3$

Average Error = 2.33 pixels

Therefore:

Average Error of Method A = 2.33 pixels

**Step 3: Calculate Average Error for Method B**

Method B errors are:

3, 5, and 4 pixels.

First, add the errors:

$$3 + 5 + 4 = 12 \text{ pixels}$$

Number of frames = 3

$$\text{Average Error} = 12 / 3$$

$$\text{Average Error} = 4.00 \text{ pixels}$$

Therefore:

$$\text{Average Error of Method B} = 4.00 \text{ pixels}$$

#### **Step 4: Compare the Results**

| Method | Average Error |
|--------|---------------|
|--------|---------------|

|          |             |
|----------|-------------|
| Method A | 2.33 pixels |
|----------|-------------|

|          |             |
|----------|-------------|
| Method B | 4.00 pixels |
|----------|-------------|

Difference in error:

$$4.00 - 2.33 = 1.67 \text{ pixels}$$

Method A has a lower average error by 1.67 pixels.

#### **(b) Evaluation**

In motion estimation, the method with the lower displacement error is considered more accurate because its estimated motion is closer to the actual motion.

Method A has an average error of 2.33 pixels, whereas Method B has an average error of 4.00 pixels.

Method A also produces lower errors in all three frames:

- Frame 1: 2 pixels < 3 pixels
- Frame 2: 3 pixels < 5 pixels
- Frame 3: 2 pixels < 4 pixels

Therefore, Method A consistently performs better.

#### **Final Conclusion**

Method A is more accurate than Method B for motion estimation. Its average displacement error is only 2.33 pixels, compared with 4.00 pixels for Method B. Since lower error indicates better accuracy, Method A is the preferred technique for the given data.

### 3. Optical Flow Magnitude Comparison

Optical flow magnitudes (in pixels/frame):

| Region | Method A | Method B |
|--------|----------|----------|
| R1     | 5        | 8        |
| R2     | 6        | 10       |
| R3     | 5        | 9        |

(a) Compute average motion magnitude for both methods.

(b) Compare and evaluate which method captures stronger motion.

#### Given Data

The optical flow magnitudes produced by two methods are:

| Region | Method A       | Method B        |
|--------|----------------|-----------------|
| R1     | 5 pixels/frame | 8 pixels/frame  |
| R2     | 6 pixels/frame | 10 pixels/frame |
| R3     | 5 pixels/frame | 9 pixels/frame  |

(a) Compute Average Motion Magnitude for Both Methods

#### Step 1: Write the formula

Average Motion Magnitude = Total Motion Magnitude / Number of Regions

A higher optical flow magnitude indicates stronger measured motion.

#### Step 2: Calculate Average Motion Magnitude for Method A

Method A values are:

5, 6, and 5 pixels/frame.

First, add the values:

$$5 + 6 + 5 = 16 \text{ pixels/frame}$$

Number of regions = 3

Average Motion Magnitude =  $16 / 3$

= 5.33 pixels/frame

Therefore:

Average Motion Magnitude of Method A = 5.33 pixels/frame

### **Step 3: Calculate Average Motion Magnitude for Method B**

Method B values are:

8, 10, and 9 pixels/frame.

First, add the values:

$8 + 10 + 9 = 27$  pixels/frame

Number of regions = 3

Average Motion Magnitude =  $27 / 3$

= 9 pixels/frame

Therefore:

Average Motion Magnitude of Method B = 9 pixels/frame

### **Step 4: Compare the Results**

| Method   | Average Motion Magnitude |
|----------|--------------------------|
| Method A | 5.33 pixels/frame        |
| Method B | 9 pixels/frame           |

Difference:

$9 - 5.33 = 3.67$  pixels/frame

Therefore, Method B has an average motion magnitude that is 3.67 pixels/frame higher than Method A.

### **(b) Evaluation**

Method B produces a higher optical flow magnitude in all three regions:

- R1: Method A = 5, Method B = 8
- R2: Method A = 6, Method B = 10



- R3: Method A = 5, Method B = 9

The average magnitude of Method B is 9 pixels/frame, while Method A is only 5.33 pixels/frame. This indicates that Method B captures stronger apparent motion across the regions.

### **Important Interpretation**

A higher optical flow magnitude means stronger measured motion, but it does not necessarily mean that the method is more accurate. To determine accuracy, actual or ground-truth motion values would be required.

### **Final Conclusion**

Method B captures stronger motion than Method A because its average optical flow magnitude is 9 pixels/frame, compared with 5.33 pixels/frame for Method A. Therefore, based on motion magnitude, Method B is the better technique for capturing stronger apparent motion.

## **4. Depth Estimation Comparison (Stereo vs SfM)**

**Estimated depth values (in meters):**

**Object Stereo Vision Structure from Motion**

**A     2.0     2.5**

**B     3.0     3.8**

**C     4.0     5.0**

**Actual depths: A=2, B=3, C=4**

**(a) Compute absolute error for both methods.**

**(b) Compare and evaluate which method is more accurate.**

**Given Data**

| <b>Object</b> | <b>Stereo Vision</b> | <b>Structure from Motion (SfM)</b> | <b>Actual Depth</b> |
|---------------|----------------------|------------------------------------|---------------------|
| A             | 2.0 m                | 2.5 m                              | 2.0 m               |
| B             | 3.0 m                | 3.8 m                              | 3.0 m               |
| C             | 4.0 m                | 5.0 m                              | 4.0 m               |

**(a) Compute Absolute Error**

**Formula:**

$$\text{Absolute Error} = |\text{Estimated Depth} - \text{Actual Depth}|$$

### **Stereo Vision**

#### **Object A:**

$$\text{Absolute Error} = |2.0 - 2.0| = \mathbf{0\ m}$$

#### **Object B:**

$$\text{Absolute Error} = |3.0 - 3.0| = \mathbf{0\ m}$$

#### **Object C:**

$$\text{Absolute Error} = |4.0 - 4.0| = \mathbf{0\ m}$$

Average Error:

$$\text{Average Error} = (0 + 0 + 0) / 3$$

$$= \mathbf{0\ m}$$

### **SfM**

#### **Object A:**

$$\text{Absolute Error} = |2.5 - 2.0| = \mathbf{0.5\ m}$$

#### **Object B:**

$$\text{Absolute Error} = |3.8 - 3.0| = \mathbf{0.8\ m}$$

#### **Object C:**

$$\text{Absolute Error} = |5.0 - 4.0| = \mathbf{1.0\ m}$$

Average Error:

$$\text{Average Error} = (0.5 + 0.8 + 1.0) / 3$$

$$= 2.3 / 3$$

$$= \mathbf{0.77 \text{ m}}$$

### **Error Comparison**

| <b>Object</b>        | <b>Stereo Error</b> | <b>SfM Error</b> |
|----------------------|---------------------|------------------|
| A                    | 0 m                 | 0.5 m            |
| B                    | 0 m                 | 0.8 m            |
| C                    | 0 m                 | 1.0 m            |
| <b>Average Error</b> | <b>0 m</b>          | <b>0.77 m</b>    |

### **(b) Evaluation and Conclusion**

Stereo Vision has an average absolute error of **0 m**, while SfM has an average error of **0.77 m**.

Since **lower error means better accuracy**, Stereo Vision performs better.

### **Final Conclusion**

Stereo Vision is more accurate than SfM because it has zero average error, whereas SfM has an average error of 0.77 m.

## **5. Pattern Recognition Accuracy**

**Comparison Two recognition systems**

**produce results:**

| <b>System</b> | <b>Correct Predictions</b> | <b>Total Samples</b> |
|---------------|----------------------------|----------------------|
|---------------|----------------------------|----------------------|

|                    |           |            |
|--------------------|-----------|------------|
| <b>Traditional</b> | <b>85</b> | <b>100</b> |
|--------------------|-----------|------------|

|                      |           |            |
|----------------------|-----------|------------|
| <b>Deep Learning</b> | <b>92</b> | <b>100</b> |
|----------------------|-----------|------------|

**(a) Compute accuracy for both systems.**

**(b) Compare and evaluate which system performs better.**

**Given Data**

| System        | Correct Predictions | Total Samples |
|---------------|---------------------|---------------|
| Traditional   | 85                  | 100           |
| Deep Learning | 92                  | 100           |

**(a) Compute Accuracy for Both Systems****Step 1: Write the formula**

$$\text{Accuracy} = (\text{Correct Predictions} / \text{Total Samples}) \times 100$$

**Step 2: Calculate Traditional System Accuracy**

$$\text{Correct Predictions} = 85$$

$$\text{Total Samples} = 100$$

$$\text{Accuracy} = (85 / 100) \times 100$$

$$\text{Accuracy} = \mathbf{85\%}$$

Therefore:

$$\text{Traditional System Accuracy} = \mathbf{85\%}$$

**Step 3: Calculate Deep Learning Accuracy**

$$\text{Correct Predictions} = 92$$

$$\text{Total Samples} = 100$$

$$\text{Accuracy} = (92 / 100) \times 100$$

$$\text{Accuracy} = \mathbf{92\%}$$

Therefore:

$$\text{Deep Learning Accuracy} = \mathbf{92\%}$$

## **(b) Compare and Evaluate Which System Performs Better**

### **Step 4: Compare the Results**

| <b>System</b>        | <b>Accuracy</b> |
|----------------------|-----------------|
| Traditional          | 85%             |
| <b>Deep Learning</b> | <b>92%</b>      |

### **Step 5: Calculate the Difference**

Difference in accuracy:

$$92\% - 85\% = 7 \text{ percentage points}$$

This means the Deep Learning system correctly predicts **7 more samples out of every 100 samples** than the Traditional system.

### **Step 6: Evaluation**

The Traditional system correctly predicts **85 out of 100 samples**, while the Deep Learning system correctly predicts **92 out of 100 samples**.

Since **higher accuracy indicates better recognition performance**, the Deep Learning system performs better.

The result shows that Deep Learning provides more accurate pattern recognition for the given dataset.

### **Final Conclusion**

Deep Learning is the better-performing system because it achieves an accuracy of 92%, compared with 85% for the Traditional system. Therefore, Deep Learning improves the recognition accuracy by 7 percentage points.